

Zoey Shu

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Skills

- **Programming languages:** Python, JavaScript, React.js, Node.js, MongoDB, HTML/CSS, Java, Flask, C++, Next.js
- **App/Web development:** Git, Docker, Google Cloud CLI, AWS, Android Studio, RESTfulAPI, FastAPI
- **ML/Data libraries:** Pytorch, Tensorflow, Scikit-learn, Pandas, Matplotlib, Seaborn, Beautiful soup

Experiences

Research Engineer

Singapore-MIT Alliance for Research & Technology Centre (Remote)

Jun - Sep 2024

- Led the system implementation of a **theory-driven** AI platform for deductive qualitative analysis. Architected a global, hierarchical data management system with **Zustand** to optimize data retrieval from user inputs and uploaded files, integrating structured prompt embeddings with LLM API interactions. Achieved a **40%** improvement in data fetching speed and reduced data loss by **20%**.
- Authored key sections of application-associated research paper, including **system architecture**, **primary features**, and **related works**. Created detailed diagrams for experiment result analysis. The paper has been submitted to CHI 2025 and is currently under review.
- Contributed to human-AI (HAI) collaboration research, annotating over **300** publications published from ACL 2021 - 2024 to classify and evaluate existing HAI interaction frameworks.

Software Engineer Intern

Nexa AI, San Jose, CA

Jan - May, 2024

- Developed a series of **RESTful API** endpoints using **Java** and **Spring Boot** to process incoming requests from the frontend.
- Implemented **Controller**, **DTOs** and **Model** layers with **Spring MVC** to manage HTTP requests and responses, optimizing data transfer between layers and improving performance. Constructed aggregation pipelines using **Spring MongoDB** and **MongoTemplate** to perform querying and manipulation in MongoDB databases.
- Developed interactive features using **React**, and **Context API** for state management. And Utilized **components**, **hooks** and **libraries** to create responsive and scalable web interfaces. Collaborated with designers to improve UI with **Ant Design** and **Tailwind CSS**, and integrated **Tableau dashboard** for data visualization.
- Led the website **SEO** enhancement solutions, and applied effective **image scaling** and **Google Cloud CDN** optimization. Achieved a **30% reduction** in search and loading time spent.
- Utilized **llama.cpp** to quantize language models on **Ollama (Long Chain)** to convert models to Q2 - F16 **GGUF** formats.
- Applied **LoRa (Keras)** for **fine-tuning** the model to facilitate cross-application collaboration tasks.
- Developed an application with **Gradios** to host the model on **Hugging Face** Spaces, enabling direct interaction with the updated models and facilitating real-time feedback. Resulting in a **33%** improvement in content generation quality and a **1.5%** decrease in memory usage. The quantized models acquired **1521** overall downloads.
- Developed a **FastAPI** platform using Python to facilitate integration between **third-party LLMs** and application APIs.
- Configured **HTTP middleware** for API security, managing request authorization headers to ensure data security.
- Engineered the **mobile** version of the platform using Java in **Android Studio**, developing essential utility class modules to enable a function-driven language model supporting on-device actions.

Projects and publications

When LLM Meets Hypergraph: A Sociological Analysis on Personality via Online Social Networks

Oct 2023 - Apr 2024

Accepted by [CIKM](#)

- Initiated research with a **Deep Hypergraph Convolutional Framework** and **LLMs** to classify users' **MBTI** personality types on social media networks. Enhancing models' **11.91%** accuracy of overall performance.

Thumb up or head down? The impact of upvote on social media self-efficacy

May - Dec 2023

Accepted by [ICSPHS Conference](#)

- Conducted therapeutic and qualitative methods to analyze the psychological impacts of users' behaviors on social media.

Education/Certification

University of California, Berkeley

Aug 2018 - Jun 2020

Master degree of landscape architecture

Berkeley, CA

Colorado State University

Aug 2014 - Aug 2018

Bachelor degree of landscape architecture

Fort Collins, CO